

Calc 2 – Taylor & Maclaurin Series

Worksheet A

1. Write the Maclaurin series for $\sin x$ through degree 7.
2. Find the Maclaurin series for $f(x) = \frac{1}{1+x^2}$.
3. Find the Taylor series of $\ln x$ centered at $a = 1$ (first four nonzero terms).
4. Use a known series to evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$.
5. Estimate $\sin(0.1)$ using two nonzero terms of its Maclaurin series; give an error bound.

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Worksheet B

1. Write the Maclaurin series for $\cos x$ through degree 6.
2. Find a Maclaurin series for xe^{-x^2} .
3. Find the Taylor series of $\sqrt{x}$ centered at $a = 4$ (first three nonzero terms).
4. Use a known series to evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.
5. Estimate $e^{0.2}$ using terms through x^3 of its Maclaurin series; give an error bound.

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Answer Key (Worksheets A & B)

Worksheet A.

1. $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots$

2. Substitute $-x^2$ into $\frac{1}{1-u} = \sum u^n$: $\boxed{\sum_{n=0}^{\infty} (-1)^n x^{2n}}$, $|x| < 1$.

3. Let $u = x - 1$. $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \frac{u^4}{4} + \dots$, so $\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots$.

4. $\cos x = 1 - \frac{x^2}{2} + O(x^4)$, so $1 - \cos x = \frac{x^2}{2} + O(x^4)$, and $\frac{1 - \cos x}{x^2} \rightarrow \boxed{1/2}$.

5. $\sin(0.1) \approx 0.1 - \frac{(0.1)^3}{6} \approx 0.099833$. Alternating series error $\leq \frac{(0.1)^5}{120} \approx 8.3 \times 10^{-8}$.

Worksheet B.

1. $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \dots$

2. $e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!}$, so $xe^{-x^2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!}$.

3. $f(x) = x^{1/2}$ at $a = 4$: $f(4) = 2$, $f'(4) = 1/4$, $f''(4) = -1/32$. $\sqrt{x} = 2 + \frac{x-4}{4} - \frac{(x-4)^2}{64} + \dots$.

4. $e^x = 1 + x + \frac{x^2}{2} + O(x^3)$, so $e^x - 1 - x = \frac{x^2}{2} + O(x^3)$, ratio $\rightarrow \boxed{1/2}$.

5. $e^{0.2} \approx 1 + 0.2 + \frac{0.04}{2} + \frac{0.008}{6} = 1.22133$. Lagrange bound $\leq \frac{e^{0.2}(0.2)^4}{24} \leq \frac{1.25 \cdot 0.0016}{24} \approx 8.3 \times 10^{-5}$.